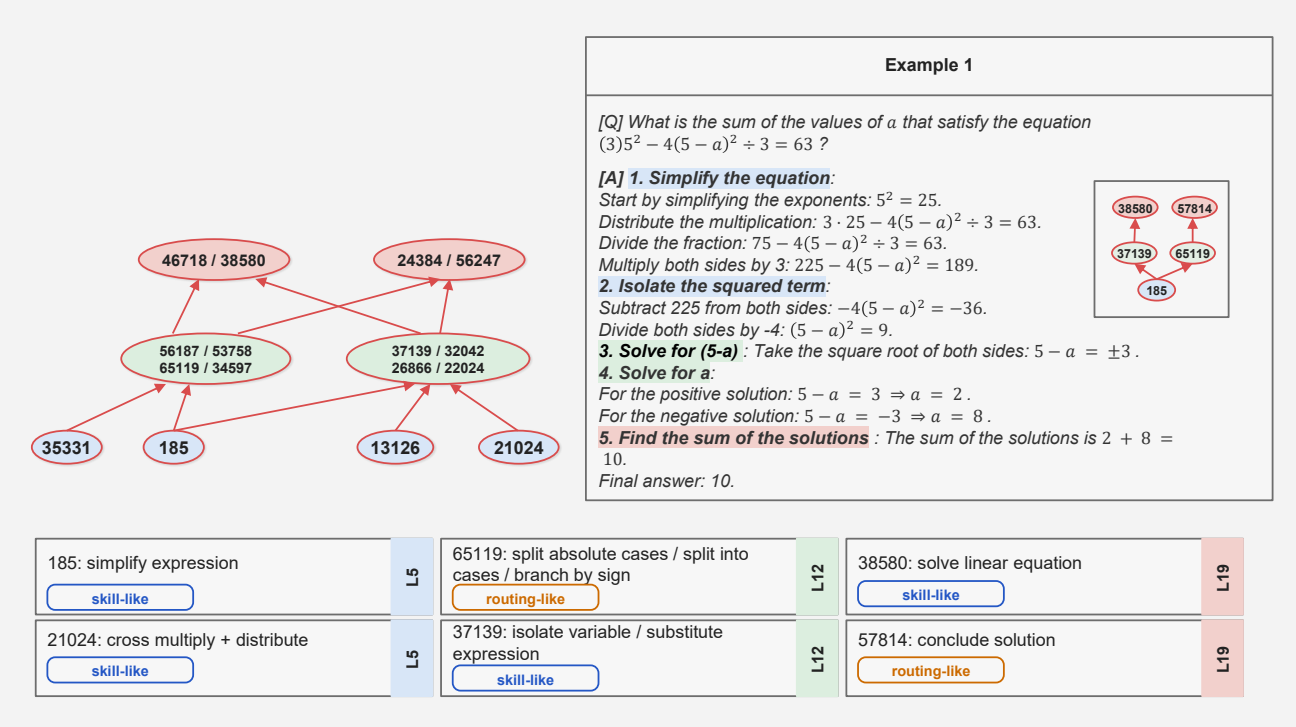

Figure H10. **The learned hierarchical concept graph for autoregressive language models.** Based on real-world math benchmarks, we applied pretrained SAEs to Gemma-2-2b-it, partitioning tokens by sequence position (early, intermediate, final), then ran causal discovery across SAE features. Both the intermediate layer and the high-level layer contain skill-like and routing-like families. The resulting graph reveals a hierarchical structure: a high-level node ("conclude solution") is caused by an intermediate node ("branch by sign"). This shows the hierarchical discrete structure our theory posits does emerge in real transformer representations.

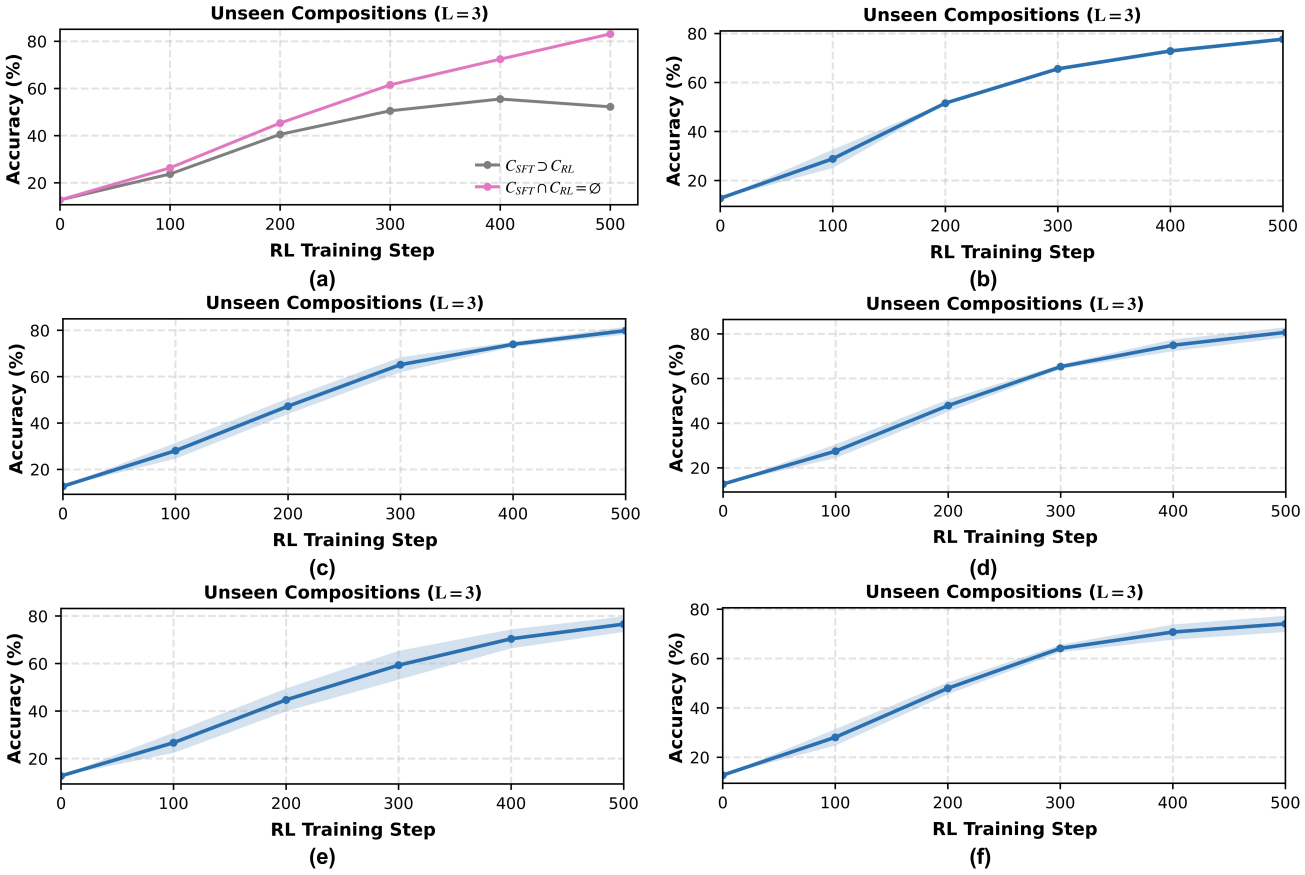

Figure H11. **The results of the full sensitivity experiments.**